Delirium screening (CAM-ICU) — Q35

Describe how you would screen him for delirium using the CAM-ICU

- Step 1: assess arousal with RASS; CAM-ICU can be done if RASS −3 to +4 (if −4/−5, reassess later)
- Feature 1: acute change or fluctuating course of mental status
- Feature 2: inattention (letters test 'SAVEAHAART' – > 2 errors)
- Feature 3: altered level of consciousness (RASS ≠ 0)
- Feature 4: disorganised thinking (yes/no questions, commands)
- Positive = 1 + 2 + (3 or 4); ICDSC ≥ 4 is an alternative tool

List the subtypes of delirium and the predisposing and precipitating factors

- Hyperactive, hypoactive (most common, worst prognosis, often missed), mixed
- Predisposing: age, dementia, alcohol, hearing/visual impairment, frailty
- Precipitating: benzodiazepines, sepsis, hypoxia, metabolic derangement, pain, sleep deprivation, restraints
- Anticholinergic drugs, urinary retention, constipation, withdrawal

What are the consequences of ICU delirium

- Increased mortality (each day of delirium raises risk)
- Longer ventilation and ICU/hospital stay
- Long-term cognitive impairment (BRAIN-ICU)
- Self-extubation, line removal, falls; higher costs

Postoperative delirium — Q36

What is the first step in managing her delirium

- Identify and treat the cause: check hypoxia, hypoglycaemia, pain, urinary retention, constipation, infection, electrolytes, drugs (opioids, anticholinergics, benzodiazepines), alcohol withdrawal
- Ensure safety; do not restrain if avoidable
- Non-pharmacological measures first (reorientation, family, glasses/hearing aids, sleep hygiene)

Describe the ABCDEF bundle

- A: assess, prevent and manage pain
- B: both spontaneous awakening and breathing trials
- C: choice of analgesia and sedation (light sedation, avoid benzodiazepines)
- D: delirium – assess, prevent, manage
- E: early mobility and exercise
- F: family engagement and empowerment

What is the role of drugs

- No drug prevents or treats delirium routinely (PADIS 2018)
- MIND-USA and AID-ICU: haloperidol does not reduce delirium duration or mortality
- Short-term low-dose antipsychotic for severe agitation endangering safety
- Dexmedetomidine for agitated delirium preventing extubation (DahLIA)

Osmotherapy and sodium target — Q37

Compare hypertonic saline and mannitol for raised ICP

- Both create an osmotic gradient drawing water from brain to plasma; mannitol also lowers viscosity
- Mannitol 0.25–1 g/kg over 15–20 min: osmotic diuresis → hypovolaemia, hypotension, AKI
- Hypertonic saline (e.g., 3% 250 mL, or 23.4% 30 mL via central line): volume expander, maintains BP/CPP
- HTS may be more effective and longer-lasting; no clear mortality difference (COBI trial: continuous HTS no benefit)

What sodium and osmolality targets would you use during osmotherapy

- Target serum Na+ 145–155 mmol/L
- Stop or hold mannitol if serum osmolality > 320 mOsm/kg
- Osmolar gap > 20 mOsm/kg indicates mannitol accumulation (risk of AKI)

List the complications of osmotherapy

- Rebound intracranial hypertension on stopping (leakage across a disrupted BBB)
- Hypernatraemia, hyperchloraemic acidosis (HTS), fluid overload, pulmonary oedema
- Mannitol: AKI, hypovolaemia, hyperkalaemia
- Osmotic demyelination if chronic hyponatraemia corrected rapidly

Osmotherapy and sodium target — Q38

Calculate her osmolar gap if Na 150 mmol/L, glucose 6 mmol/L and urea 10 mmol/L, and interpret it

- Calculated osmolality = 2 Na + glucose + urea = 300 + 6 + 10 = 316 mOsm/kg
- Osmolar gap = measured − calculated = 335 − 316 = 19 mOsm/kg
- Gap near 20 suggests mannitol accumulation (not cleared)
- Risk of mannitol-induced AKI → stop mannitol, switch to hypertonic saline if needed

Explain how mannitol causes AKI

- Osmotic nephrosis: vacuolisation of proximal tubular cells
- Renal vasoconstriction at high concentrations
- Hypovolaemia from osmotic diuresis

How would you manage her fluids now

- Replace urine losses with isotonic fluid to avoid hypovolaemia
- Keep Na 145–155 with hypertonic saline if ICP control still needed
- Monitor electrolytes (K, Mg, PO4) and urine output closely
- Haemodialysis if severe AKI with high osmolar gap (removes mannitol) – use slow/CRRT to avoid ICP rise

Tiered management of raised ICP — Q39

Describe the Tier 0 and Tier 1 measures of the SIBICC algorithm

- Tier 0 (baseline): head up 30°, neutral neck, ETT ties loose; sedation/analgesia; normothermia; Na > 135; SpO2 ≥ 94%, PaCO2 35–45; Hb > 7; MAP for CPP ≥ 60
- Tier 1: CPP 60–70 mmHg, deepen analgesia/sedation
- Tier 1: intermittent mannitol or hypertonic saline boluses
- Tier 1: CSF drainage via EVD if present; EEG for seizures; consider anticonvulsants

Describe the Tier 2 and Tier 3 measures

- Tier 2: mild hypocapnia (PaCO2 32–35 mmHg), neuromuscular blockade trial, MAP challenge to assess autoregulation, raise CPP if autoregulation intact
- Tier 3: secondary decompressive craniectomy
- Tier 3: barbiturate (thiopental/pentobarbital) coma titrated to burst suppression
- Tier 3: mild hypothermia (35–36 °C) – Eurotherm showed harm from early routine use

Which measures should NOT be used routinely

- Steroids (CRASH: increased mortality in TBI)
- Prophylactic hypothermia; profound hyperventilation (PaCO2 < 30)
- Routine CSF lumbar drainage
- Routine CPP > 90 mmHg

Tiered management of raised ICP — Q40

Explain the Monro–Kellie doctrine and intracranial compliance

- Skull is a fixed box: brain + blood + CSF volume is constant
- A rise in one must be compensated by a fall in another (CSF shift to spinal canal, venous blood displaced)
- When compensation is exhausted, small volume changes cause large ICP rises (steep part of the pressure–volume curve)

Summarise the evidence for decompressive craniectomy in TBI

- DECRA (2011): early bifrontal DC for moderate ICP rise – lower ICP, but worse functional outcome
- RESCUEicp (2016): DC for refractory ICP > 25 – lower mortality (27% vs 49%), more survivors in vegetative state/severe disability
- At 24 months more favourable outcomes with DC in RESCUEicp
- Decision must include discussion of likely disability with family

Describe the ICP waveform and what a P2 > P1 means

- P1 percussion wave (arterial pulse), P2 tidal wave (brain compliance), P3 dicrotic wave (aortic valve closure)
- Normally P1 > P2 > P3
- P2 > P1 indicates reduced intracranial compliance

Malignant MCA infarction (HAMLET) — Q41

Define malignant MCA infarction and its features

- Large MCA infarct (> 50% of MCA territory, or DWI volume > 145 mL) with space-occupying oedema
- Progressive decline in consciousness within 24–72 h; herniation
- Mortality up to 80% with medical treatment alone

Summarise the evidence for decompressive hemicraniectomy (DESTINY, DECIMAL, HAMLET)

- Pooled analysis (≤ 60 years, surgery within 48 h): mortality reduced from ~71% to ~22%
- More patients with mRS ≤ 3 (moderate disability) at 1 year
- HAMLET: benefit if surgery within 48 h; no benefit if delayed up to 96 h
- DESTINY II (> 60 years): improves survival, but most survivors severely disabled (mRS 4)

Outline the medical management while awaiting surgery

- Head up 30°, normothermia, normoglycaemia, treat seizures
- Osmotherapy (mannitol or HTS) as a bridge
- Airway protection if GCS ≤ 8
- BP management: avoid hypotension; ≤ 185/110 if thrombolysis given

Malignant MCA infarction (HAMLET) — Q42

What factors would influence the decision for hemicraniectomy

- Age (strongest evidence ≤ 60 years), time from onset (≤ 48 h)
- Pre-stroke function and comorbidities
- Patient's previously stated values about survival with disability
- Dominant hemisphere (aphasia) – historically a concern, but outcomes similar; surgery should not be denied on side alone

How would you counsel the family

- Surgery greatly improves the chance of survival
- Most survivors, especially > 60 years, will remain dependent (mRS 4: unable to walk without help)
- Ask what she would have wanted; shared decision-making; time-limited trial if uncertain

What are the complications of decompressive craniectomy

- Infection, haematoma, CSF leak
- Hydrocephalus, subdural hygroma
- 'Syndrome of the trephined' (sinking skin flap)
- Need for later cranioplasty and its complications

Status epilepticus — Q43

Define status epilepticus using the ILAE time points

- t1 = 5 min for convulsive SE: time after which seizure is unlikely to stop on its own → start treatment
- t2 = 30 min: time after which long-term neuronal injury is likely

Outline the stepwise treatment with drugs and doses

- 0–5 min: ABC, O2, glucose check, IV access, thiamine if alcohol/malnutrition
- 5–20 min: benzodiazepine – lorazepam 0.1 mg/kg IV (max 4 mg, repeat once) or IM midazolam 10 mg
- 20–40 min (established SE): levetiracetam 60 mg/kg IV (max 4.5 g) or fosphenytoin 20 mg PE/kg or valproate 40 mg/kg – all equally effective (ESETT)
- 40+ min (refractory SE): intubate; anaesthetic infusion – midazolam, propofol or thiopental
- Look for the cause: CT, electrolytes, toxicology, LP if infection suspected

What are the systemic complications of prolonged convulsive status epilepticus

- Hypoxia, aspiration, airway obstruction
- Lactic acidosis, hyperthermia, rhabdomyolysis → AKI
- Arrhythmias, neurogenic pulmonary oedema
- Hyperglycaemia then hypoglycaemia; cerebral oedema and neuronal injury

Status epilepticus — Q44

What is your next step and how would you titrate therapy

- Refractory SE → continuous anaesthetic infusion
- Midazolam (0.2 mg/kg bolus, 0.05–2 mg/kg/h) or propofol (1–2 mg/kg, 2–5 mg/kg/h – watch PRIS) or thiopental/pentobarbital
- Titrate to seizure suppression or burst suppression on continuous EEG for 24–48 h, then wean slowly
- Add another second-line drug (e.g., valproate or lacosamide); vasopressors often needed

Define super-refractory status epilepticus and list the treatment options

- SE continuing ≥ 24 h after starting anaesthesia, or recurring on weaning
- Options: ketamine (NMDA antagonist), barbiturates, additional AEDs (perampanel, topiramate)
- Immunotherapy (steroids, IVIG, plasma exchange) for autoimmune causes; ketogenic diet; hypothermia

What is non-convulsive status epilepticus and how is it diagnosed

- Ongoing electrographic seizures without obvious motor activity
- Suspect in coma, fluctuating consciousness, subtle twitching, after convulsive SE
- Diagnosis needs EEG (Salzburg criteria)

Apnoea test for brain death — Q45

List the prerequisites before brain death testing

- Known, irreversible cause of catastrophic brain injury (neuroimaging consistent)
- Exclusion of confounders: sedatives (wait ≥ 5 half-lives or levels), neuromuscular blockade (train-of-four)
- Core temperature ≥ 36 °C; SBP ≥ 100 mmHg or MAP ≥ 60 mmHg
- No severe electrolyte, acid–base or endocrine disturbance

Describe how the apnoea test is performed and the criteria for a positive test

- Pre-oxygenate with FiO2 1.0 for ≥ 10 min; baseline ABG with PaCO2 35–45 mmHg, normal pH
- Disconnect (or CPAP), give O2 via catheter to the carina (6 L/min) or CPAP 10 cmH2O
- Observe for any respiratory movement for 8–10 min
- Positive (consistent with brain death): no breathing with PaCO2 ≥ 60 mmHg AND ≥ 20 mmHg above baseline (pH < 7.30)
- Abort if SBP < 90 mmHg, SpO2 < 85% or arrhythmia → ancillary test

When are ancillary tests needed

- Apnoea test cannot be completed safely
- Confounders cannot be excluded; facial/eye injury prevents examination
- High cervical cord injury
- Tests: CT angiography, cerebral perfusion scintigraphy, four-vessel angiography, EEG

Apnoea test for brain death — Q46

Explain the physiological changes after brainstem death

- Catecholamine storm (Cushing reflex) → hypertension, tachycardia, myocardial injury, neurogenic pulmonary oedema
- Then loss of sympathetic tone → vasodilatation and hypotension
- Diabetes insipidus (posterior pituitary failure) → polyuria, hypernatraemia
- Hypothermia, coagulopathy (tissue factor release), hypothyroidism, adrenal insufficiency

Outline the donor management targets

- MAP ≥ 60–70 mmHg; vasopressin first-line (also treats DI), then noradrenaline
- Desmopressin for DI; Na 135–150 mmol/L; replace urine losses
- Lung-protective ventilation (VT 6–8 mL/kg, PEEP 5–10), recruitment, SpO2 > 92%
- Methylprednisolone 15 mg/kg; consider T3/T4; glucose 4–10 mmol/L; temperature > 36 °C

What are the ethical principles of organ donation after brain death

- Dead donor rule
- Separation of the death-confirming team from the transplant team
- Consent/authorisation from the patient's prior wishes or family
- Optimise organs without causing harm to the dignity of the patient

Aneurysmal SAH and nimodipine — Q47

What drug improves neurological outcome after aneurysmal SAH? Give the regimen

- Nimodipine 60 mg orally (or NG) every 4 hours for 21 days
- If hypotension: 30 mg every 2 hours
- Improves outcome by reducing delayed cerebral ischaemia, not angiographic vasospasm

Describe your assessment of her new deficit

- Likely delayed cerebral ischaemia (DCI) – peaks day 4–14
- Exclude other causes: hydrocephalus, rebleeding, seizures, hyponatraemia, hypoxia, infection, hypoglycaemia
- Urgent CT and CT angiography/perfusion; transcranial Doppler (mean velocity > 120 cm/s, Lindegaard ratio > 3)
- Check sodium, glucose, ABG and EEG if NCSE suspected

Outline the management of DCI and treatments that are NOT recommended

- Euvolaemia; induced hypertension (raise SBP in steps with vasopressor) while monitoring response
- Endovascular rescue: intra-arterial vasodilators (verapamil, milrinone, nimodipine) or angioplasty if refractory
- Not recommended: prophylactic triple-H therapy (hypervolaemia harmful)
- Not recommended: routine magnesium (MASH-2), statins (STASH), clazosentan for outcome

Aneurysmal SAH and nimodipine — Q48

Grade his SAH using WFNS and modified Fisher scales

- WFNS grade II (GCS 13–14 without motor deficit)
- Modified Fisher grade 4 (thick SAH with IVH)
- Modified Fisher 3–4 indicates highest risk of vasospasm/DCI

Describe the early management until the aneurysm is secured

- Secure aneurysm as early as possible (within 24 h); endovascular coiling preferred if suitable (ISAT)
- SBP < 160 mmHg (reduce rebleeding risk)
- Nimodipine; analgesia; antiemetics; treat seizures if present (no routine prophylaxis)
- EVD if hydrocephalus/IVH with reduced consciousness
- Short course tranexamic acid not recommended routinely (ULTRA)

Compare SIADH and cerebral salt wasting after SAH

- Both: hyponatraemia with concentrated urine (high urine Na)
- SIADH: euvolaemic/hypervolaemic → fluid restriction is NOT advised in SAH (risk of DCI)
- CSW: hypovolaemic (natriuresis + polyuria) → isotonic/hypertonic saline, fludrocortisone